Consistent but Blind?

Image removal does not measure image reliance in medical vision-language models

Tri Thai Huu · VinUniversity, Hanoi, Vietnam · thaitri2005@gmail.com

paper, code, data

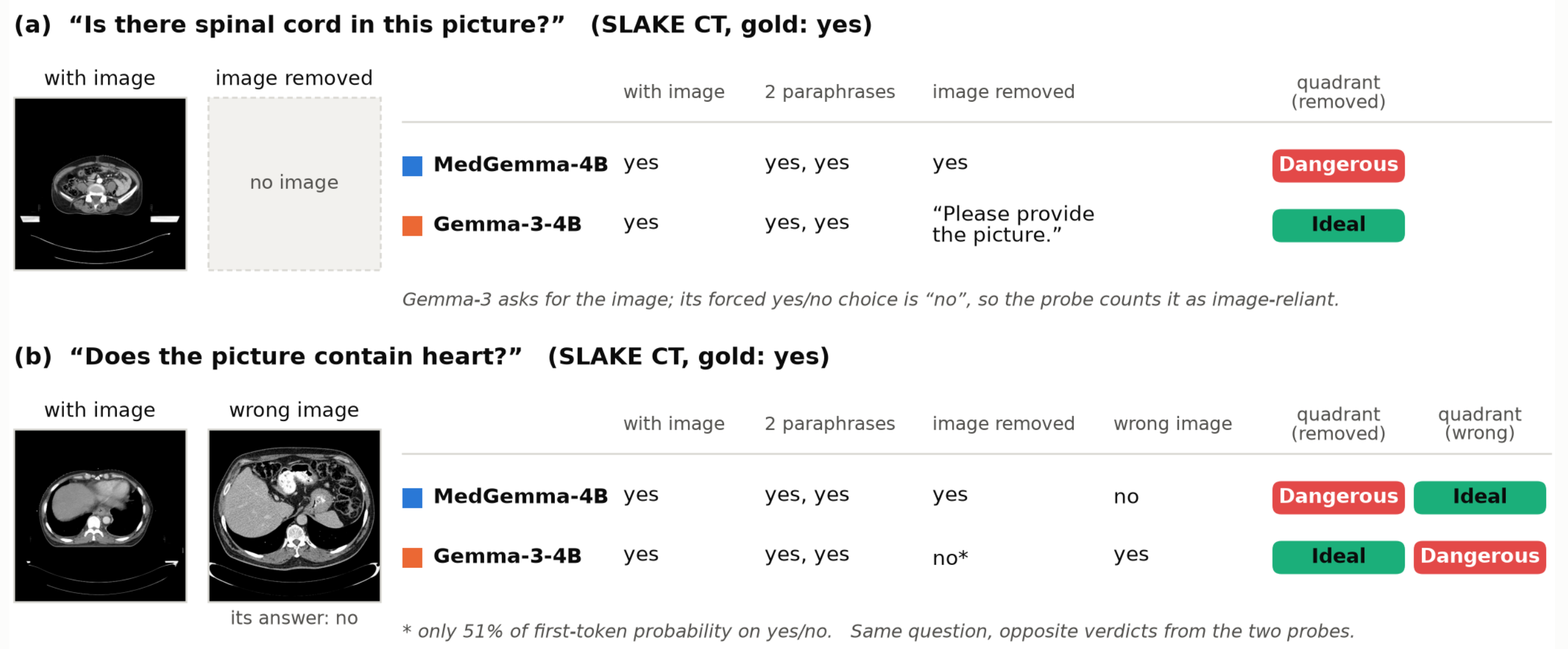
The problem

A medical VLM can answer correctly from priors without looking at the image. The *consistent-but-dangerous* taxonomy flags such answers:

	changes without image	**same** without image
stable under paraphrase	Ideal	Dangerous
flips under paraphrase	Fragile	Worst

Its reliance test is **removing the image**. Can it tell us **which model** relies on the image more?

Two models, one question



(a) "Is there spinal cord in this picture?" (SLAKE CT, gold: yes)

	with image	2 paraphrases	image removed	quadrant (removed)
MedGemma-4B	yes	yes, yes	yes	Dangerous
Gemma-3-4B	yes	yes, yes	"Please provide the picture."	Ideal

Gemma-3 asks for the image; its forced yes/no choice is "no", so the probe counts it as image-reliant.

(b) "Does the picture contain heart?" (SLAKE CT, gold: yes)

	with image	2 paraphrases	image removed	wrong image	quadrant (removed)	quadrant (wrong)
MedGemma-4B	yes	yes, yes	yes	no	Dangerous	Ideal
Gemma-3-4B	yes	yes, yes	no*	yes	Ideal	Dangerous

** only 51% of first-token probability on yes/no. Same question, opposite verdicts from the two probes.*

Without the image MedGemma answers (Dangerous) and Gemma-3 asks for the image (Ideal). A wrong image gives the opposite verdict.

Contributions

- Four reliance probes compared on the same questions, 3 medical/general pairs.
- What image removal measures instead: **refusal** and **priors**.
- A pre-registered test **outside the training data**.

Method

Three pairs, each a medical model and its general base:
MedGemma-4B, **MedGemma-1.5-4B** vs. **Gemma-3-4B**
MedVLThinker-3B vs. **Qwen2.5-VL-3B**

Data: SLAKE (CT, MRI, X-ray), VQA-RAD, PathVQA; 3,365 questions, 907 images.

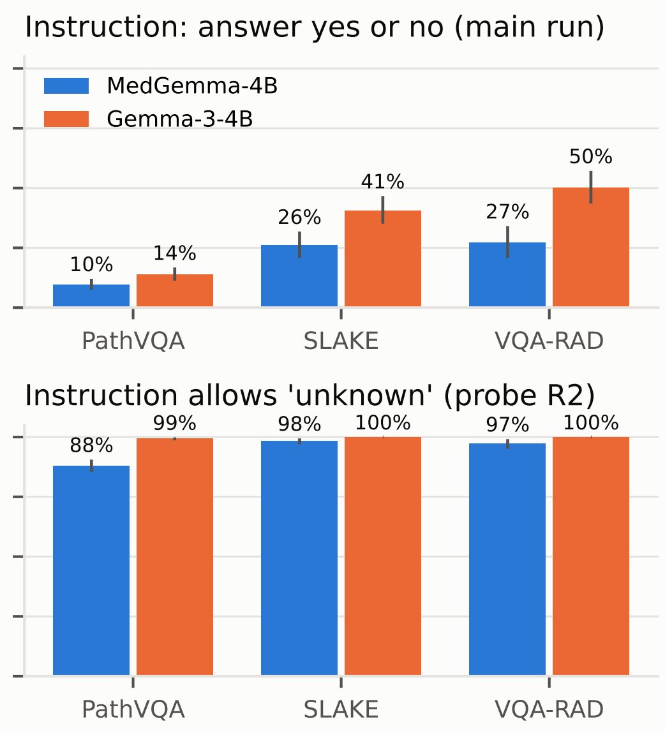
Probes change only the image; we count how often the yes/no answer changes.

Image removed	image tokens omitted (the taxonomy's test)
Wrong image	another image, same modality
Black image	black image in the image slot
Opposite-answer	same question, an image where the answer is the opposite
"Unknown" allowed	image removed, may decline

Scoring: P(yes) from the yes/no token logits in one forward pass; the yes+no mass detects refusals. bf16, greedy, identical prompts; about 120,000 calls on one 16 GB GPU.

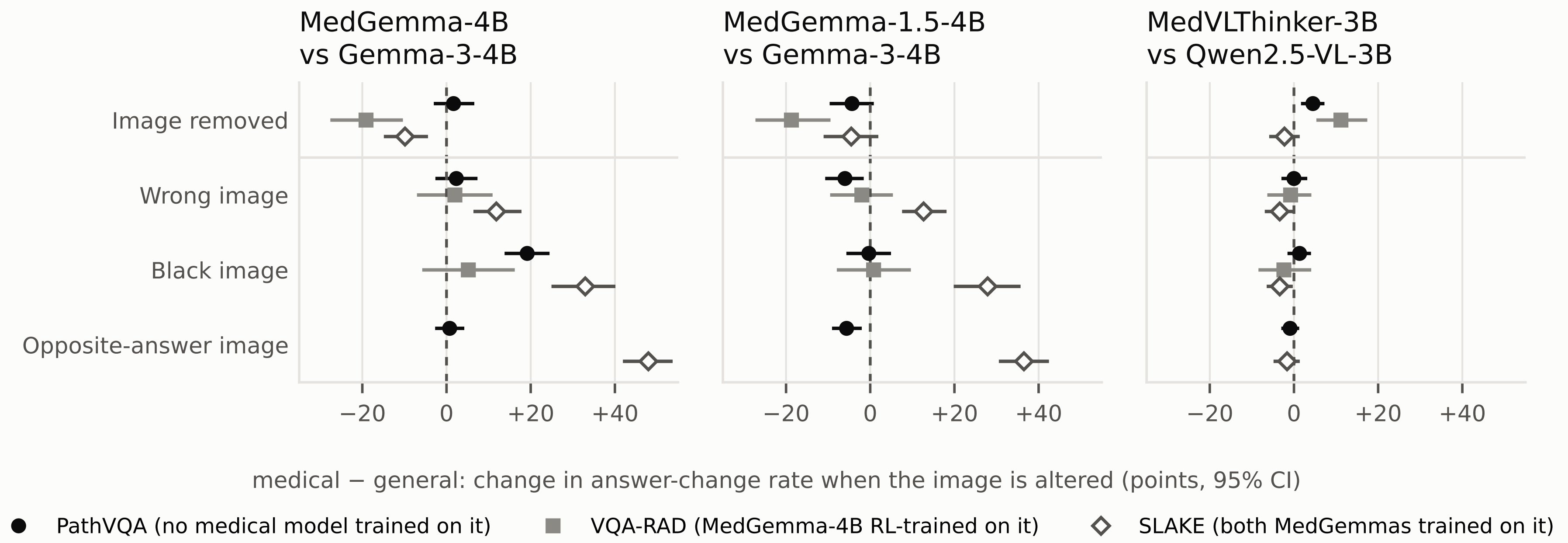
Pre-registered: endpoints frozen before the main run; each later probe and model pair registered (dated git tag) before it ran. 95% CIs by image.

Who answers without the image?



Share of questions declined without the image. Forced to say yes or no (top), Gemma-3 declines far more; allowed to say "unknown" (bottom), both decline on $\geq$93%.

Finding 1: image removal agrees with no in-format probe



Medical – general difference in how often a yes/no answer changes when the image is altered.

5 of 9
cells where image removal separates the models

0 of 5
of those confirmed by an in-format probe

+13.7
points more Dangerous for MedGemma-4B, image removed

Why: what image removal picks up

Refusal. Without an image, **Gemma-3** asks for it on **29%** of questions, **MedGemma-4B** on 18%, **MedGemma-1.5** on 0.7%. Its forced answers there are unrelated to its image answers, so they count as "image-reliant".

Priors. A blind guess that matches the image leaves the answer unchanged, so image removal calls it blind. MedGemma's extra Dangerous answers are **all correct** (+14.1 correct, −0.4 wrong).

Finding 2: the in-format advantage stays in the training data

Follows the opposite-answer image, medical – general (points):

	SLAKE trained on	**PathVQA** held out
MedGemma-4B	+48.0	+0.8
MedGemma-1.5	+36.5	–5.6
MedVLThinker	–1.6	–0.9

The PathVQA test was **registered before it ran**. The MedGemma predictions failed, so "MedGemma uses the image more" was **withdrawn**.

The Dangerous share, corrected

- 36–42% of yes/no answers look image-independent **by chance**.
- Its excess is mostly **correct** answers.
- Reliance flags errors at chance (AUROC **0.51**).

Take-aways

1. Don't rank models by image removal.
2. Probe in format, with an opposite-answer image.
3. Test outside each model's training data.

Limits

Three pairs; the explanations are post hoc; the held-out test is pathology only; paraphrases LLM-judged (AI-assisted 50-item check: 88%). Not a clinical claim.

Sadanandan & Behzadan, CVPR-W 2026; Zafar et al. 2026; Lotfinia et al. 2026; Sellergren et al., MedGemma, 2025.